

José Arbelaez

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EDUCATION

Universidad Autónoma de Madrid | BSc in Data Science and Engineering 2022 – 2026

- Received highest honors in Natural Language Processing, Computer Vision, Computer Networks, High-Performance Computing, Statistical Methods and Machine Learning.
- Board Member of the university Computer Security Club (SEIF UAM), 2023 – 2024.

Google Cloud | Cloud & Data Engineering Certificate 2025 – Present

EXPERIENCE

Naudit HPCN | Data Scientist Intern Jan 2026 – May 2026

- Managed Logstash, Filebeat and Elastic Agent pipelines processing more than 50 TB of banking-platform data per day, correcting production errors and monitoring the system with Grafana and Prometheus.
- Developed an alarm autocorrelation system using 17 statistical features, calibrated HistGradientBoosting models, SHAP explanations and causal DAG visualization, achieving F1 above 0.80.

Google Developer Group on Campus UAM | President & Co-Founder Jul 2025 – Present

- Co-founded the chapter, coordinate a multidisciplinary team and lead AI, cloud and software-development events for more than 200 students; designed gdguam.es and helped implement several of its components.

VULCASAN SL | Full-Stack Developer Jun 2024 – Jul 2025

- Built a production application automating client, inventory and operational workflows for an automotive workshop network; developed backend APIs, authentication and CI/CD in a four-person team.

SELECTED PROJECTS

Bachelor's Thesis | Surrogate Models [\[Read article\]](#) 2025 – 2026

- Built a Gaussian-process and Expected-Improvement framework for UAM's Healthy Food Ingredients Group to prioritize untested *Hermetia illucens* diets from scarce data, reducing the search space for costly food-biology experiments.

RoadGuard | Reliable Road-Damage Detection [\[Read article\]](#) 2026

- Turned crack and pothole detection into a deployment-oriented inspection system with cross-country validation, calibrated confidence, abstention and explainable maintenance prioritization.

BGG Review Intelligence | NLP & Opinion Mining [\[Read article\]](#) 2025

- Built a reproducible NLP pipeline that converts unstructured BoardGameGeek reviews into sentiment predictions and aspect-level evidence for product and player-experience decisions.

Image Generation with Diffusion Models [\[Read article\]](#) 2025

- Implemented VP, VE and Sub-VP diffusion models from scratch on CIFAR-10, covering score matching, multiple samplers, conditional generation and image imputation.

HONORS & AWARDS

- **1st of 240+ participants – UAM Investors League**; achieved the competition's highest Sharpe ratio with a risk-adjusted portfolio strategy, UAM and IronIA Fintech (2025).
- **2nd Place – Cloud Agent Systems Hackathon**, Google Cloud and Diverger (2025).
- **Academic Excellence Scholarship**, Community of Madrid (2024 and 2025).

LANGUAGES & TECHNICAL SKILLS

- **Languages:** Spanish – Native; English – C1; French – Basic.
- **AI/ML:** Scikit-learn, PyTorch, TensorFlow/Keras, Gaussian processes, Bayesian optimization, SHAP, computer vision and NLP.
- **Data & Infrastructure:** PostgreSQL, DuckDB, MongoDB, Elasticsearch, Logstash, Grafana, Prometheus, Docker, Linux, Git and GCP.

ADDITIONAL INFORMATION

- **Interests & Activities:** Open-source contributions, building practical hardware and automation for my home garden, and playing the harmonica.